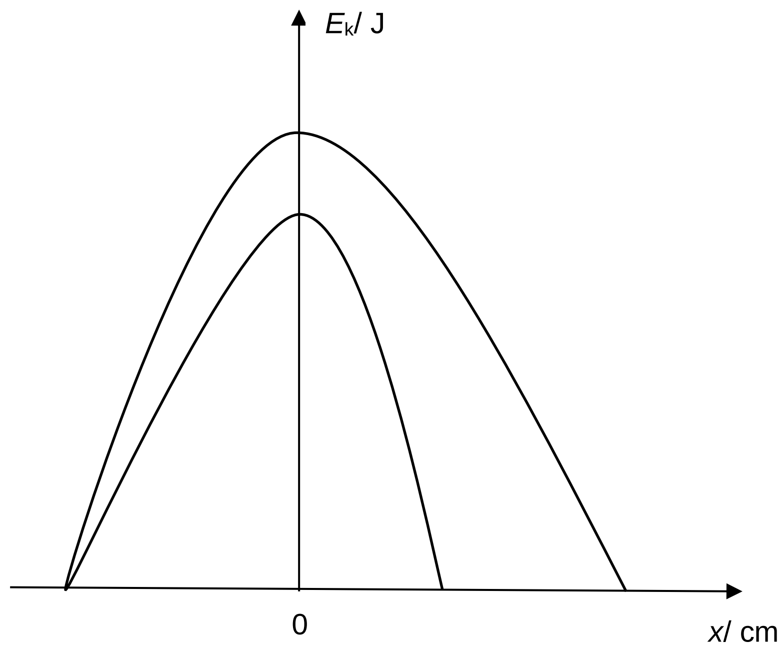
Question				Solution	Marks
7	**(a)**	**(i)**		g and R are constants, hence a is proportional to x the negative sign indicates that a is in the opposite direction of x.	**B1** **B1**
		(ii)		$\omega = 2\pi / T$ and $T = 2.2$ s $\omega = 2\pi / 2.2 = 2.9$ rad s^{-1}	 **A1**
		(iii)		$\omega^2 = g / R$ $R = 9.81 / 2.86^2$ $= 1.2$ m	**A1**
		(iv)		$v = \omega\sqrt{x_o^2 - x^2}$ $v = (2.9)\sqrt{(0.030)^2 - (0.015)^2}$ $= 0.075$ m s^{-1}	**C1** **A1**
		(v)		Total energy = maximum KE $= \frac{1}{2} m v_0^2$ $= \frac{1}{2} m(\omega x_0)^2$ $= \frac{1}{2}(0.045)(2.9 \times 3.0 \times 10^{-2})^2$ $= 1.7 \times 10^{-4}$ J	**B1** **M1** **A1**
		(vi)		$E_k = \frac{1}{2} m v_0^2 \sin^2 \omega t$ $E_k = 1.7 \times 10^{-4} \sin^2(2.9t)$	**C1** **A1**
		(vii)		 *To show amplitude and kinetic energy decreasing clearly*	**B1**
	(b)	**(i)**	**1.**	$v = f\lambda$ $0.90 = f(0.30)$ $f = 3.0$ Hz	 **A1**
			2.	$3.0 = \frac{1}{2\pi}\sqrt{\frac{28}{m}}$ $m = 0.079$ kg	 **A1**
		(ii)	**1.**	(Since energy is proportional to amplitude squared) more energy is being transferred,	**M1** **A1**

				Amplitude increases	
			2.	Frequency of wave decreases. Does not match natural frequency Amplitude decreases	**M1** **A1**
			3.	Mass of system increases. Natural frequency drops. Does not match driver frequency of the waves. Amplitude decreases.	**M1** **A1**